

System 12: Distributed Search Engine & Real-Time Typeahead (Elasticsearch Architecture)

Zero-Prerequisite Intuition: The "Index at the Back of the Book" Metaphor Why can't a normal SQL database handle text search? Imagine you have a 1,000-page textbook on Medicine, and you want to find every page that mentions the word *"Penicillin"*. If you use a standard SQL database query:

[Redacted SQL Query]

The database has to read word #1 on page 1, word #2 on page 1, all the way through page 1,000. That is a **Sequential Table Scan**, and on 100 million documents it takes **45 seconds!**

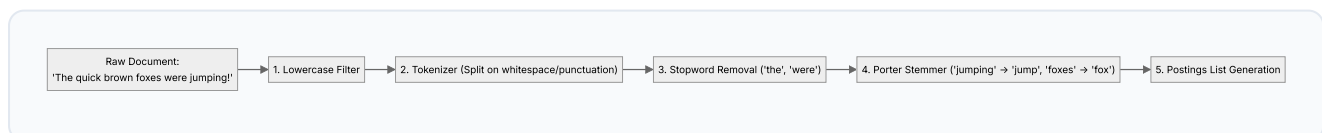
What do human readers do instead? You flip to the **very back of the book** to the alphabetical **Index section**. You look up the letter **P**, find *"Penicillin"*, and see: [Page 42, Page 108, Page 780]. You found all three occurrences in **2 seconds!**

In computer science, this is called an **Inverted Index**. Instead of mapping Document → Words, it maps Word → List of Document IDs.

This chapter covers how distributed search engines (Elasticsearch, Apache Lucene) build, compress, shard, and query inverted indexes at scale, and how to build sub-10ms **Real-Time Typeahead / Autocomplete**.

1. The Anatomy of an Inverted Index

When a new document arrives, it passes through an **Analysis & Tokenization Pipeline**:



The Postings List

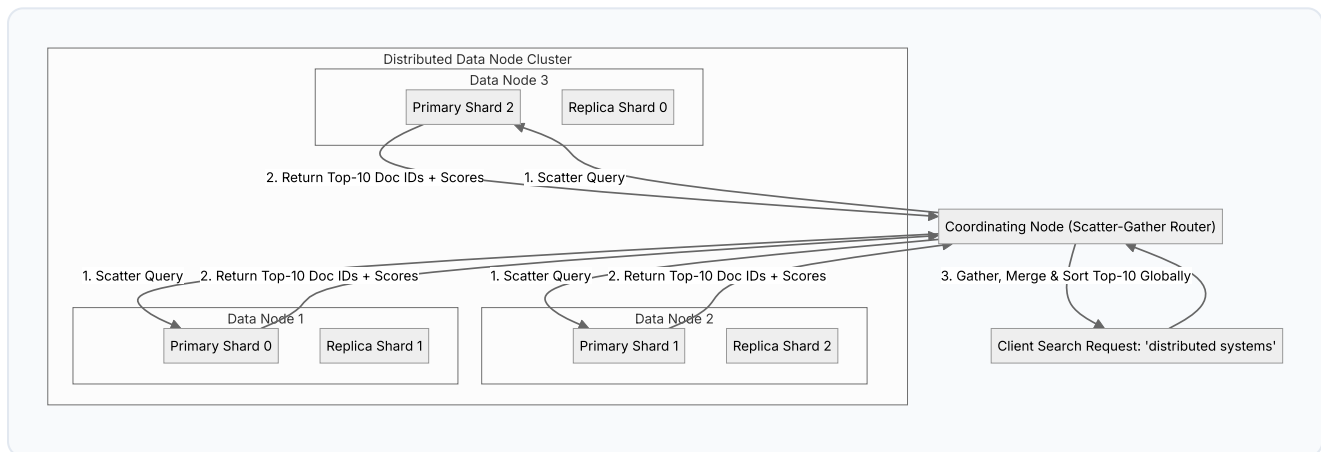
The core data structure is the **Term Dictionary** paired with **Postings Lists**:

Term (Token)	Document Frequency	Postings List (Doc IDs & Term Frequency)
brown	2	[Doc 1 (pos 2), Doc 8 (pos 14)]
fox	3	[Doc 1 (pos 3), Doc 4 (pos 1), Doc 9 (pos 5)]
jump	2	[Doc 1 (pos 5), Doc 4 (pos 2)]

When a user searches for `"brown AND jump"`, the search engine doesn't scan text—it performs an ultra-fast **Bitwise Intersection** of the two sorted postings lists:

$$\text{Postings}(\text{brown}) \cap \text{Postings}(\text{jump}) = [1, 8] \cap [1, 4] = [\mathbf{Doc\ 1}]$$

2. Distributed Elasticsearch Architecture

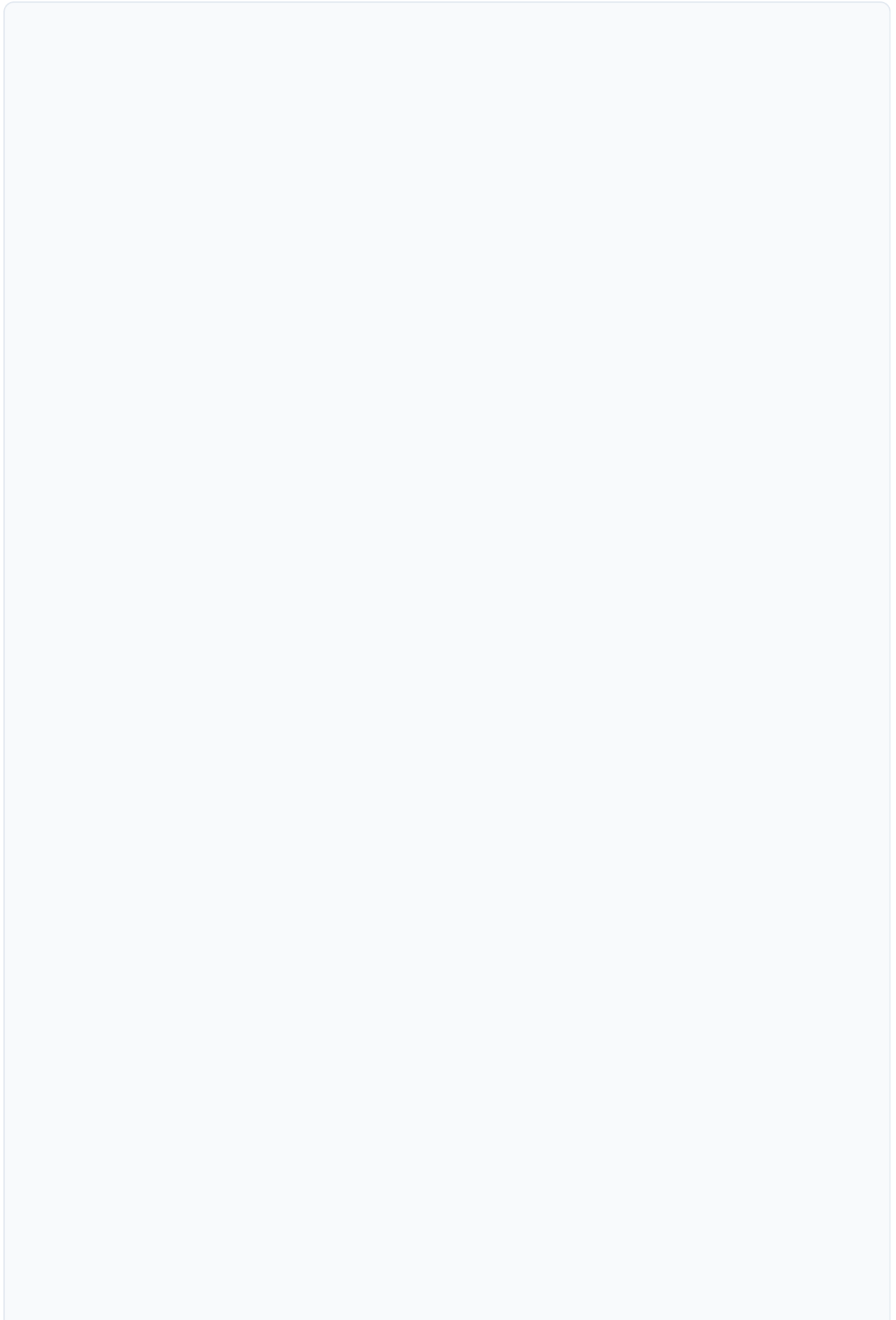


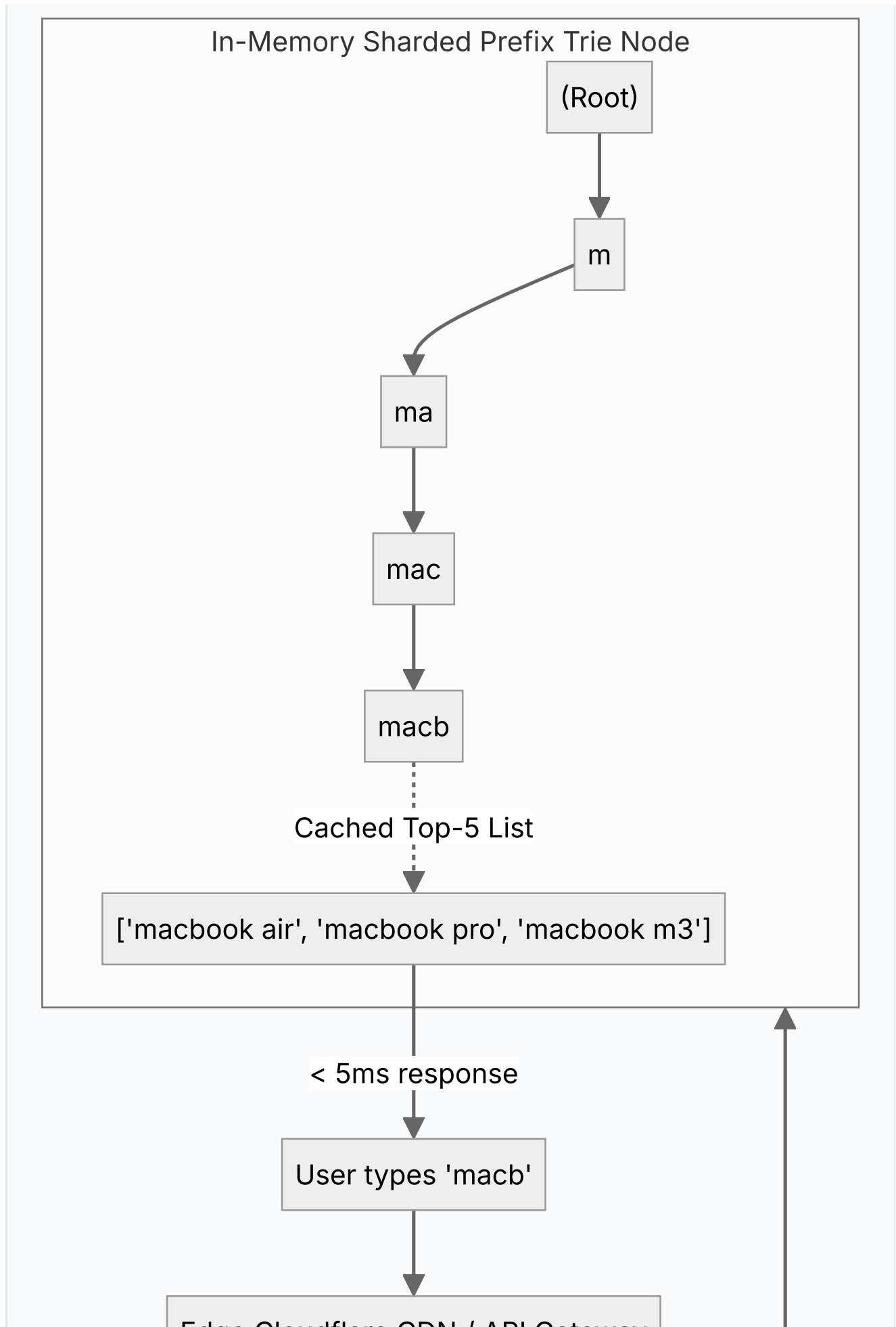
The Scatter-Gather Query Protocol:

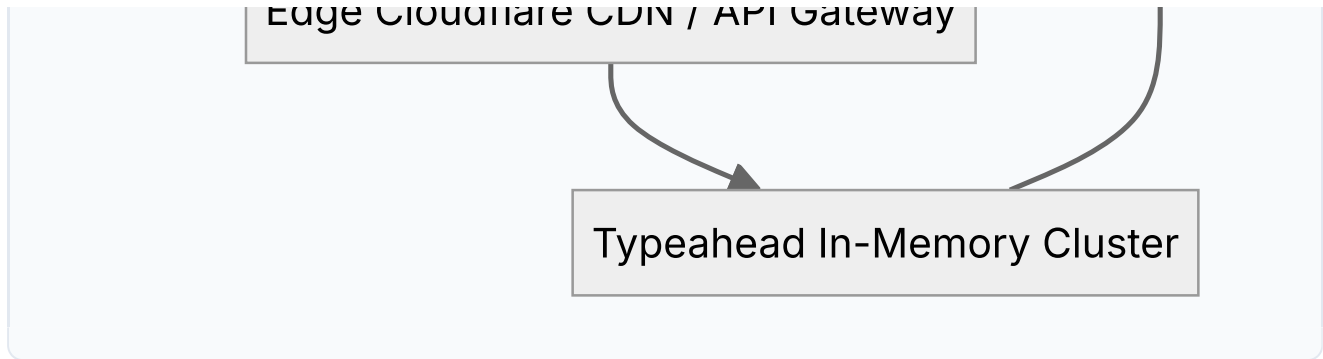
- 1. Query Phase:** The Coordinating Node hashes or broadcasts the query to one replica of every primary shard. Each shard searches its local Lucene inverted index and computes BM25 relevance scores, returning only its **Top-10 Document IDs**.
- 2. Fetch Phase:** The Coordinating Node merges the results from all shards, sorts them to pick the true global Top-10, and issues direct document retrieval requests (`GET /doc/{id}`) to fetch the full JSON bodies.

3. Real-Time Typeahead / Autocomplete Architecture

How does Google or Amazon suggest results within **8 milliseconds** as you type each letter into the search bar?







The Prefix Trie with Pre-Computed Top-K Heaps

Traversing a trie on every keystroke to find all matching children is too slow under 500,000 QPS.

- **The Optimization:** Every node in the Prefix Trie **pre-computes and caches the Top-5 most popular search terms** that pass through it!
- When the user types `m-a-c-b`, the algorithm does not search deeper—it immediately reads the cached Top-5 array stored directly on node `macb` in $O(L)$ time (where L is the length of the prefix, e.g., 4 operations)!

4. Relevance Scoring: The BM25 Formula

Modern search uses **Okapi BM25**, which balances Term Frequency (TF) against Inverse Document Frequency (IDF):

$$\text{Score}(D, Q) = \sum_{i=1}^N \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)}$$

- **Term Frequency Saturation (k_1):** If a document mentions "python" 50 times, it is not 50x more relevant than a document mentioning it 5 times. BM25 flattens the curve.
- **Document Length Normalization (b):** Prevents long 500-page documents from dominating short 1-page articles simply because they have more total words.

5. Runnable Model: Inverted Index, BM25 and Scatter-Gather

The final assertion is deliberately weak, and it points at a real distributed-search problem: **each shard computes IDF from its own documents**, so scores from different shards are not strictly comparable. Production systems either send global term statistics with the query (a

`dfs_query_then_fetch` style search) or accept small ranking differences when shards are large and evenly populated.

6. Failure Modes and Mitigations

Failure	Effect	Mitigation
Shard replica lost	Reduced capacity, possible data loss if all copies go	At least one replica per shard on a different node and zone; automatic re-replication
Slow shard	The slowest shard sets query latency	Timeouts with partial results, adaptive replica selection, hedged requests
Hot shard (skewed routing)	One node saturates	Route by a high-cardinality key; split the shard; custom routing only when needed
Mapping explosion (dynamic fields)	Memory exhausted	Strict mappings, field limits
Bulk reindex	Cluster load spikes	Index into a new index and switch an alias; throttle
Split brain	Two masters accept writes	Quorum-based master election with a majority of master-eligible nodes

7. Trade-offs and Alternatives

- **Near-real-time versus immediate visibility:** new documents become searchable after a refresh (about one second by default); forcing a refresh per write destroys indexing throughput.
- **More shards versus fewer:** many small shards parallelise but add per-shard overhead and merge cost; size shards in the tens of gigabytes.
- **Search engine versus database:** a database is the source of truth; the search index is a derived, rebuildable view, populated through change streams or a periodic reindex.
- **Relevance tuning versus simplicity:** start with BM25 and field boosts; add learning-to-rank only when you have click data to train and evaluate with.

8. Interview Timeline (45 minutes) and Follow-ups

Minutes	Do
0 to 5	Scope: corpus size, query rate, freshness, features (filters, facets, typeahead)
5 to 10	Estimates: documents, index size, queries per second
10 to 20	Inverted index, analysis (tokenising), sharding and replicas
20 to 32	Query path: scatter-gather, scoring (BM25), top-k merge
32 to 40	Indexing pipeline, near-real-time refresh, reindexing without downtime
40 to 45	Failure modes, relevance evaluation, typeahead

Follow-ups to prepare: Why are shard-local IDF scores a problem? How do you reindex a billion documents without downtime? How do you keep the search index consistent with the primary database? How would you evaluate whether a ranking change is better?